

Program 6:

Deploy an Automated Build Pipeline using Docker Hub

Step 1: Create Project Directory

```
mkdir program6
```

```
cd program6
```

Step 2: Create Node.js Application Files

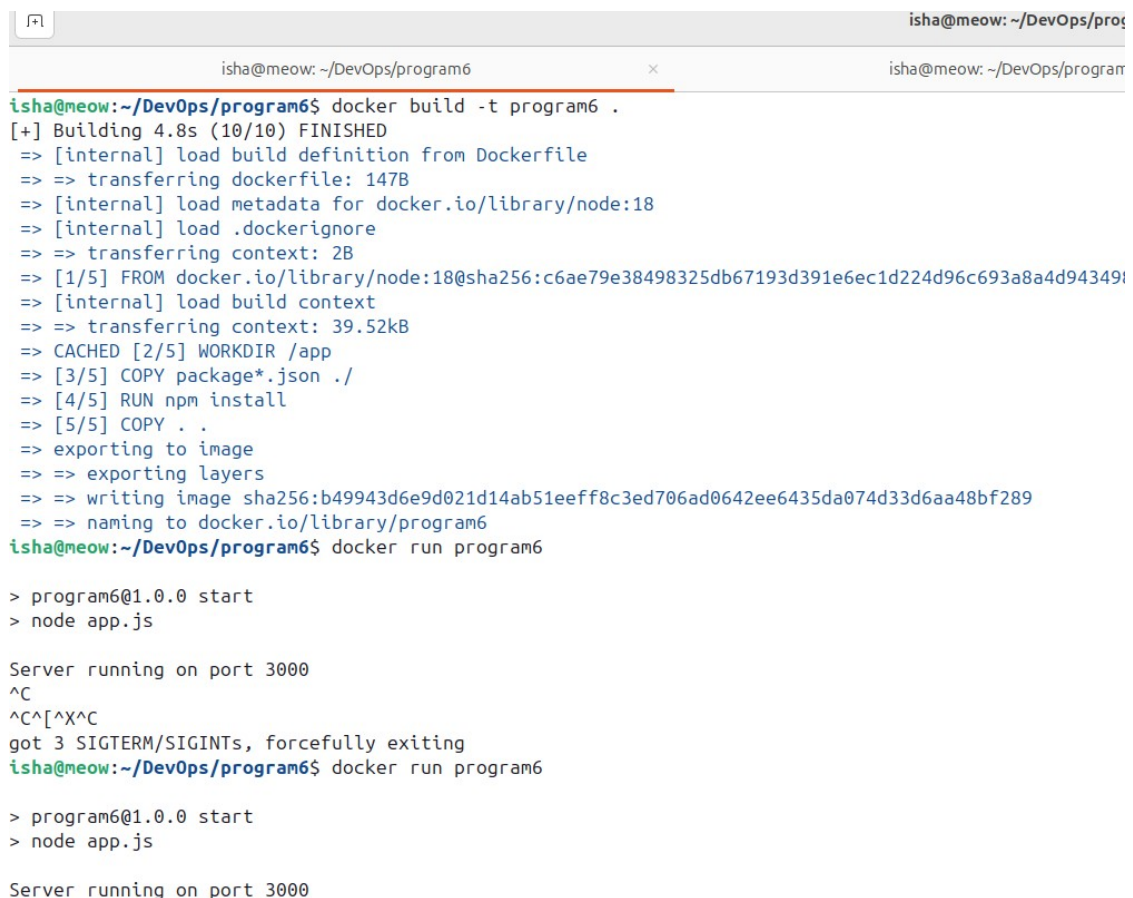
Create package.json with a start script and dependencies.

Create app.js containing a simple Express server.

Step 3: Create Dockerfile

Step 4: Build Docker Image

```
docker build -t program6 .
```



```
isha@meow: ~/DevOps/program6
isha@meow: ~/DevOps/program6$ docker build -t program6 .
[+] Building 4.8s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 147B
=> [internal] load metadata for docker.io/library/node:18
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943491
=> [internal] load build context
=> => transferring context: 39.52kB
=> CACHED [2/5] WORKDIR /app
=> [3/5] COPY package*.json ./
=> [4/5] RUN npm install
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:b49943d6e9d021d14ab51eeff8c3ed706ad0642ee6435da074d33d6aa48bf289
=> => naming to docker.io/library/program6
isha@meow: ~/DevOps/program6$ docker run program6

> program6@1.0.0 start
> node app.js

Server running on port 3000
^C
^C[^X^C
got 3 SIGTERM/SIGINTs, forcefully exiting
isha@meow: ~/DevOps/program6$ docker run program6

> program6@1.0.0 start
> node app.js

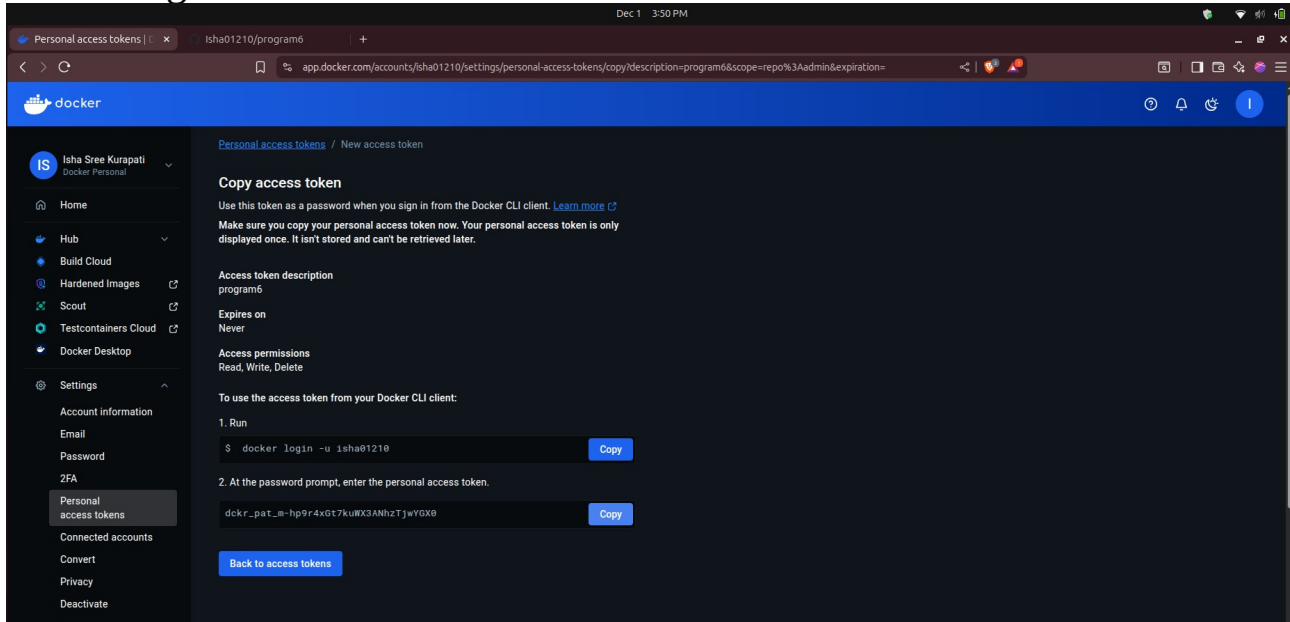
Server running on port 3000
```

Step 5: Run Docker Container Locally

`docker run program6`

Step 6: Login to Docker Hub Using Access Token

`docker login -u isha01210`



```
isha@meow: ~/DevOps
isha@meow: ~/DevOps/program6
isha@meow:~/DevOps/program6$ docker login -u isha01210
```

Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit <https://app.docker.com/settings>

Password:

WARNING! Your credentials are stored unencrypted in '/home/isha/.docker/config.json'.
Configure a credential helper to remove this warning. See
<https://docs.docker.com/go/credential-store/>

Login Succeeded

```
isha@meow:~/DevOps/program6$ docker push isha01210/program6:latest
The push refers to repository [docker.io/isha01210/program6]
ece9fe0b9046: Pushed
5a8317affd34: Pushed
1d33755c6f67: Pushed
983ea16fb306: Pushed
d2a991bcab4d: Pushed
b624aa2d5ea2: Pushed
d399c9dc306f: Pushed
84f9fa179c1b: Pushed
ce84ba212e49: Pushing [=====] 473.4MB/587.7MB
e4dc8cd9ecc8: Pushed
6428cc293366: Pushed
2f7436e79a0b: Pushed
```

Step 7: Tag Docker Image for Docker Hub

`docker tag program6 isha01210/program6:latest`

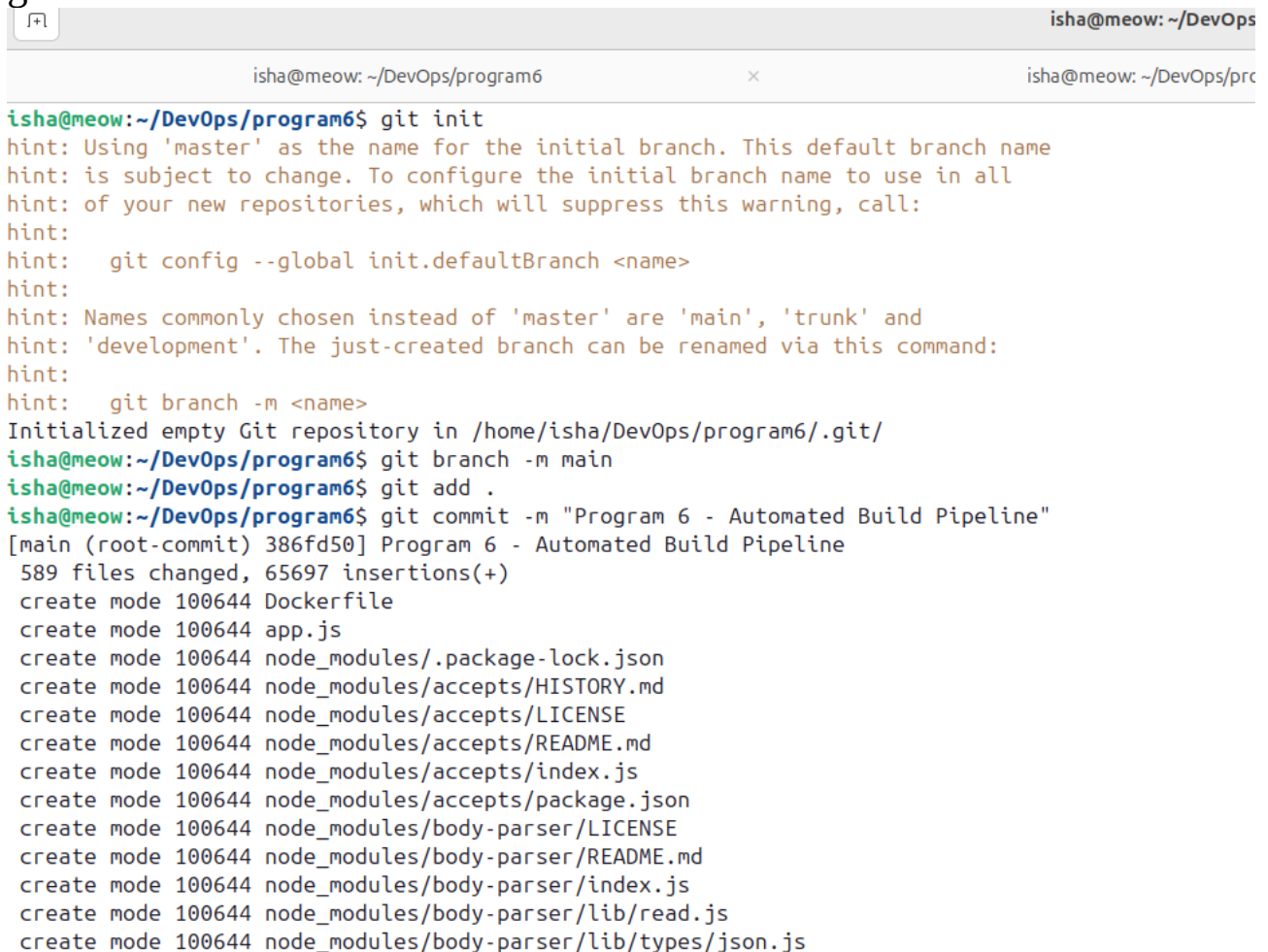
Step 8: Push Docker Image to Docker Hub

`docker push isha01210/program6:latest`

Step 9: Initialize Git Repository and rename the branch as main

`git init`

`git branch -m main`

A terminal window screenshot showing the steps to initialize a Git repository and create a commit. The terminal has a title bar with 'isha@meow: ~/DevOps' and a tab labeled 'isha@meow: ~/DevOps/program6'. The output shows the execution of 'git init', 'git branch -m main', 'git add .', and 'git commit -m "Program 6 - Automated Build Pipeline"', along with various hints and file status changes.

```
isha@meow:~/DevOps/program6$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/isha/DevOps/program6/.git/
isha@meow:~/DevOps/program6$ git branch -m main
isha@meow:~/DevOps/program6$ git add .
isha@meow:~/DevOps/program6$ git commit -m "Program 6 - Automated Build Pipeline"
[main (root-commit) 386fd50] Program 6 - Automated Build Pipeline
 589 files changed, 65697 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 app.js
 create mode 100644 node_modules/.package-lock.json
 create mode 100644 node_modules/accepts/HISTORY.md
 create mode 100644 node_modules/accepts/LICENSE
 create mode 100644 node_modules/accepts/README.md
 create mode 100644 node_modules/accepts/index.js
 create mode 100644 node_modules/accepts/package.json
 create mode 100644 node_modules/body-parser/LICENSE
 create mode 100644 node_modules/body-parser/README.md
 create mode 100644 node_modules/body-parser/index.js
 create mode 100644 node_modules/body-parser/lib/read.js
 create mode 100644 node_modules/body-parser/lib/types/json.js
```

Step 10: Add GitHub Remote Repository

git remote add origin https://github.com/isha01210/program6.git

Step 11: Stage Project Files

git add .

```
isha@meow:~/DevOps/program6$ git remote add origin https://github.com/Isha01210/program6.git
isha@meow:~/DevOps/program6$ git push -u origin main
Username for 'https://github.com': Isha01210
Password for 'https://Isha01210@github.com':
remote: Repository not found.
fatal: repository 'https://github.com/Isha01210/program6.git/' not found
isha@meow:~/DevOps/program6$ git push -u origin main
Username for 'https://github.com': Isha01210
Password for 'https://Isha01210@github.com':
Enumerating objects: 677, done.
Counting objects: 100% (677/677), done.
Delta compression using up to 8 threads
Compressing objects: 100% (626/626), done.
Writing objects: 100% (677/677), 698.07 KiB | 5.50 MiB/s, done.
Total 677 (delta 118), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (118/118), done.
To https://github.com/Isha01210/program6.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
isha@meow:~/DevOps/program6$
```

Step 12: Commit Source Code

git commit -m "Program 6 - Automated Build Pipeline"

Step 13: Push Source Code to GitHub

git push -u origin main

Step 14: Check if the repository is on git

Personal access tokens | DocIsha01210/program6

github.com/Isha01210/program6

Isha01210 / program6

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main1 Branch0 Tags

Go to file

Add file

Code

Isha01210

Program 6 - Automated Build Pipeline

386fd50 · 6 minutes ago

1 Commit

node_modules	Program 6 - Automated Build Pipeline	6 minutes ago
Dockerfile	Program 6 - Automated Build Pipeline	6 minutes ago
app.js	Program 6 - Automated Build Pipeline	6 minutes ago
package-lock.json	Program 6 - Automated Build Pipeline	6 minutes ago
package.json	Program 6 - Automated Build Pipeline	6 minutes ago

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